

## Lab 1

WAP to take input any number and throw exception if the number is even or divisible by 3.

```
import java.util.Scanner;

class NumberException extends Exception {
    NumberException(String msg) {
        super(msg);
    }
}

public class NumberCheck {
    static void checkNumber(int n) throws NumberException {
        if (n % 2 == 0)
            throw new NumberException(n + " is even.");
        else if (n % 3 == 0)
            throw new NumberException(n + " is divisible by 3.");
        else
            System.out.println(n + " is valid (odd and not divisible by 3).");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        try {
            checkNumber(num);
        } catch (NumberException e) {
            System.out.println("Exception caught: " + e.getMessage());
        }
        sc.close();
    }
}
```

### Output:

```
Enter a number: 8
Exception caught: 8 is even.
```

```
Enter a number: 9
Exception caught: 9 is divisible by 3.
```

```
Enter a number: 7
7 is valid (odd and not divisible by 3).
```

## Lab 2

*WAP to illustrate the concept of Keyboard Events.*

```
import java.awt.*;
import java.awt.event.*;

public class KeyboardEventDemo extends Frame implements KeyListener {
    Label label;

    KeyboardEventDemo() {
        label = new Label("Press any key...");
        label.setBounds(50, 100, 300, 30);
        add(label);
        addKeyListener(this);

        setSize(400, 300);
        setLayout(null);
        setVisible(true);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });
    }

    public void keyPressed(KeyEvent e) {
        label.setText("Key Pressed: " + e.getKeyChar());
    }

    public void keyReleased(KeyEvent e) {
        label.setText("Key Released: " + e.getKeyChar());
    }

    public void keyTyped(KeyEvent e) {}

    public static void main(String[] args) {
        new KeyboardEventDemo();
    }
}
```

## Lab 3

Write a code to design a form containing three text fields and two buttons "SUBMIT" and "RESET" using AWT.

```
import java.awt.*;
import java.awt.event.*;

public class FormDemo extends Frame implements ActionListener {
    TextField t1, t2, t3;
    Button submit, reset;

    FormDemo() {
        setLayout(new FlowLayout());

        add(new Label("Field 1:")); t1 = new TextField(15); add(t1);
        add(new Label("Field 2:")); t2 = new TextField(15); add(t2);
        add(new Label("Field 3:")); t3 = new TextField(15); add(t3);

        submit = new Button("SUBMIT");
        reset = new Button("RESET");
        add(submit);
        add(reset);

        submit.addActionListener(this);
        reset.addActionListener(this);

        setTitle("Form Demo");
        setSize(400, 200);
        setVisible(true);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == submit) {
            System.out.println("Submitted: " + t1.getText() + ", " +
                t2.getText() + ", " + t3.getText());
        } else if (e.getSource() == reset) {
            t1.setText(""); t2.setText(""); t3.setText("");
        }
    }

    public static void main(String[] args) {
        new FormDemo();
    }
}
```

## Lab 4

WAP to show communication between client1 and client2 using TCP.

```
import java.io.*;
import java.net.*;

public class Server {
    public static void main(String[] args) throws IOException {
        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Waiting for clients...");

        Socket s1 = ss.accept();
        System.out.println("Client1 connected");
        Socket s2 = ss.accept();
        System.out.println("Client2 connected");

        DataInputStream in1 = new DataInputStream(s1.getInputStream());
        DataOutputStream out2 = new DataOutputStream(s2.getOutputStream());
        DataInputStream in2 = new DataInputStream(s2.getInputStream());
        DataOutputStream out1 = new DataOutputStream(s1.getOutputStream());

        new Thread(() -> {
            try {
                String msg;
                while ((msg = in1.readUTF()) != null)
                    out2.writeUTF("Client1: " + msg);
            } catch (IOException e) {}
        }).start();

        new Thread(() -> {
            try {
                String msg;
                while ((msg = in2.readUTF()) != null)
                    out1.writeUTF("Client2: " + msg);
            } catch (IOException e) {}
        }).start();
    }
}

import java.io.*;
import java.net.*;

public class Client1 {
    public static void main(String[] args) throws IOException {
        Socket s = new Socket("localhost", 5000);
        DataOutputStream out = new DataOutputStream(s.getOutputStream());
        DataInputStream in = new DataInputStream(s.getInputStream());
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        new Thread(() -> {
            try {
                while (true) System.out.println(in.readUTF());
            } catch (IOException e) {}
        }).start();

        String msg;
        while ((msg = br.readLine()) != null)
            out.writeUTF(msg);
    }
}

import java.io.*;
import java.net.*;

public class Client2 {
    public static void main(String[] args) throws IOException {
        Socket s = new Socket("localhost", 5000);
        DataOutputStream out = new DataOutputStream(s.getOutputStream());
```

```
DataInputStream in = new DataInputStream(s.getInputStream());
BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

new Thread(() -> {
    try {
        while (true) System.out.println(in.readUTF());
    } catch (IOException e) {}
}).start();

String msg;
while ((msg = br.readLine()) != null)
    out.writeUTF(msg);
}
```

## Lab 5

WAP to connect MySQL database in JAVA.

```
import java.sql.*;

public class MySQLConnect {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "yourpassword";

        try {
            Connection con = DriverManager.getConnection(url, user, password);
            System.out.println("Connected to database successfully!");

            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM students");

            while (rs.next()) {
                System.out.println(rs.getInt(1) + " " + rs.getString(2));
            }

            con.close();
        } catch (SQLException e) {
            System.out.println("Connection failed: " + e.getMessage());
        }
    }
}
```